UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

9145626131

MATHEMATICS (SYLLABUS D) **4024/01**

Paper 1 **May/June 2009**

2 hours

Candidates answer on the Question Paper.

Additional Materials: Geometrical instruments

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

If working is needed for any question it must be shown in the space below that question.
Omission of essential working will result in loss of marks.

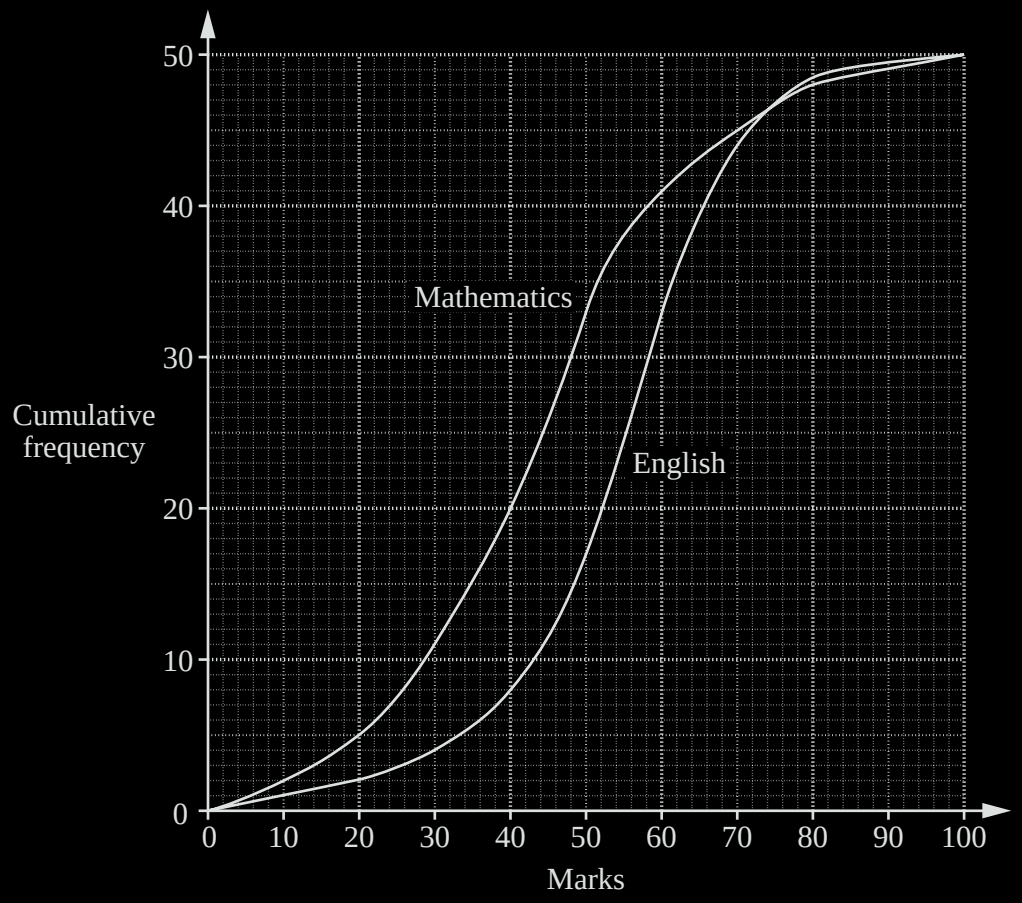
NEITHER ELECTRONIC CALCULATORS NOR MATHEMATICAL TABLES MAY BE USED IN THIS PAPER.

The number of marks is given in brackets [ ] at the end of each question or part question.
The total of the marks for this paper is 80.

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Fifty students each took a Mathematics and an English test. The distributions of their marks are shown in the cumulative frequency graph.

(a) Use the graph

(i) to estimate the median mark in the English test,

Answer (*a*)(i) .....[1]

(ii) to estimate the 20th percentile mark in the Mathematics test.

Answer (*a*)(ii) .....[1]

(b) State, **with a reason**, which test the students found more difficult.

Answer (*b*) .....

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.....

..... [1]





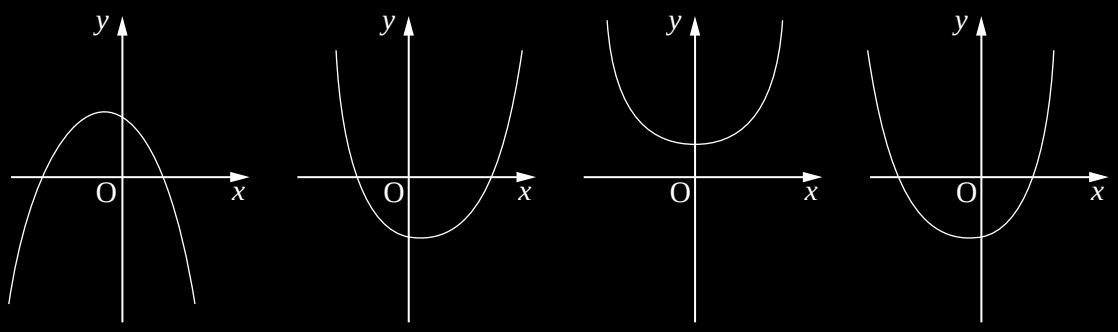


Figure 1 Figure 2 Figure 3 Figure 4

Which of the figures shown above could be the graph of

(a) $y = x^2 + 2$,

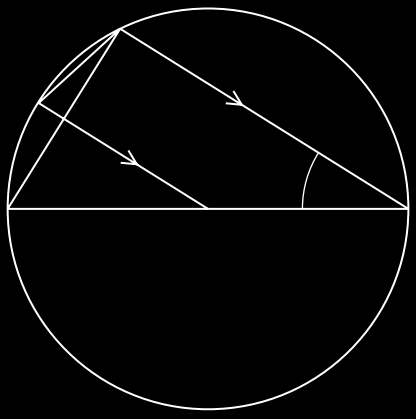
Answer (*a*) Figure .....[1]

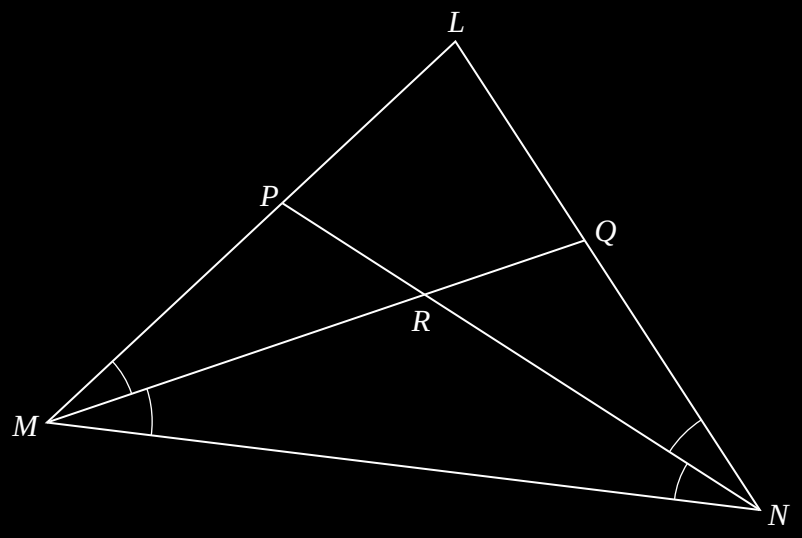
(b) $y = (x - 2)(x + 1)$,

Answer (*b*) Figure .....[1]

(c) $y = 2 - x - x^2$?

Answer (*c*) Figure.....[1]





In the diagram, $L\hat{M}Q = Q\hat{M}N = M\hat{N}P = P\hat{N}L$.

(a) Show that triangles LMQ and LNP are congruent.

[3]

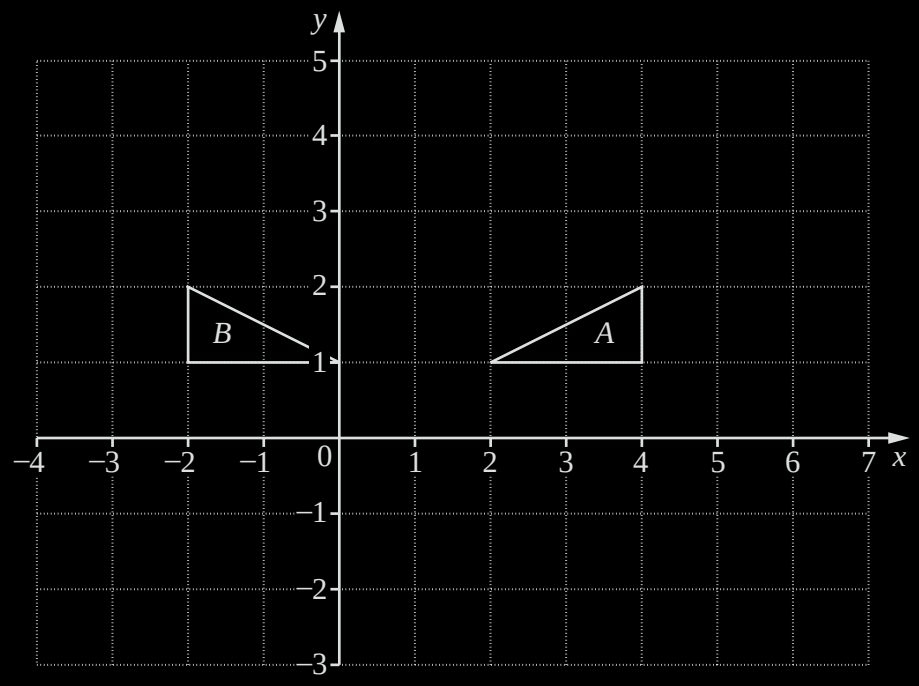
(b) Show that $M\hat{P}N = M\hat{Q}N$.

[1]

(c) The straight lines MQ and NP intersect at R.

State the name of the special quadrilateral $LPRQ$.

Answer (*c*) .....[1]



The diagram shows triangles *A* and *B*.

(a) The translation $\begin{pmatrix} -3 \\ 2 \end{pmatrix}$ maps ΔA onto ΔC.

On the diagram, draw and label ΔC. [1]

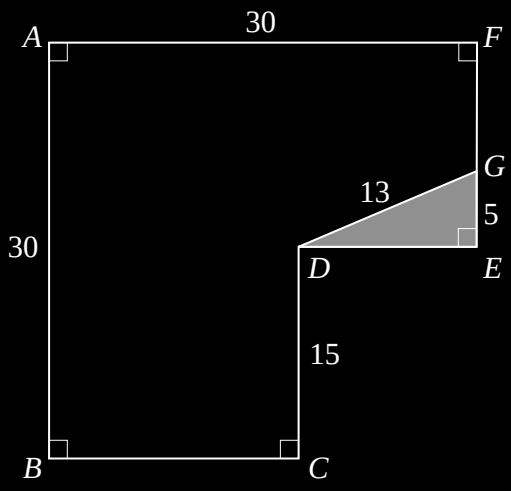
(b) The rotation 90° clockwise, centre (2, 0), maps ΔA onto ΔD.

On the diagram, draw and label ΔD. [2]

(c) Describe **fully** the **single** transformation which maps ΔA onto ΔB.

Answer (c) .....

..... [2]



ABCDEF represents an L-shaped piece of glass with $AB = AF = 30$ cm and $CD = 15$ cm.
The glass is cut to fit the window in a door and the shaded triangle *DEG* is removed.
$DG = 13$ cm and $EG = 5$ cm.

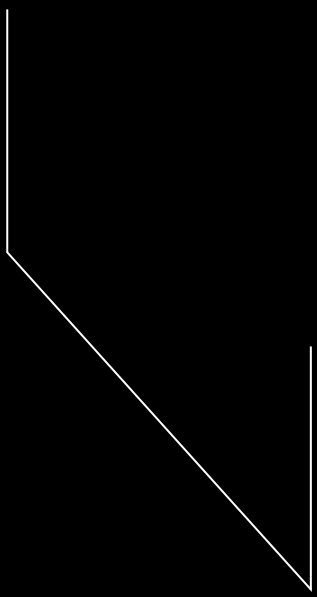
(a) Show that DE = 12 cm.

Answer (*a*) .....

.....

.....

..... [1]



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